BBACVS · BLOCKCHAIN-BASED ACADEMIC CREDENTIAL VERIFICATION SYSTEM

How a credential moves through BBACVS, and what actually lands on the blockchain

A walk through the as-built system: where each piece of data lives, which steps write to Ethereum, which stay off-chain, and why a verifier can trust the answer.

Network **Ethereum Sepolia (chain id 11155111)** Live at **bbacvs.kalootech.com** Written **15 September 2026**

Registry today **5 issuers on-chain · 12 anchored credentials · 5 ZAQA-validated**

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1 The three stores and their trust ranking

BBACVS keeps data in three places with a strict hierarchy. The blockchain is the only source that decides whether a credential is genuine. IPFS holds the encrypted document the hash was computed from. MongoDB is a convenience index for portals and workflows, and nothing in it can make a forged credential verify.

AUTHORITATIVE

Ethereum Sepolia

- Which wallet addresses are authorised issuers
- SHA-256 hash of every issued credential
- keccak hash of its IPFS content ID
- Issuer address, DID, block timestamp
- Revocations with reason code

DECENTRALISED STORE

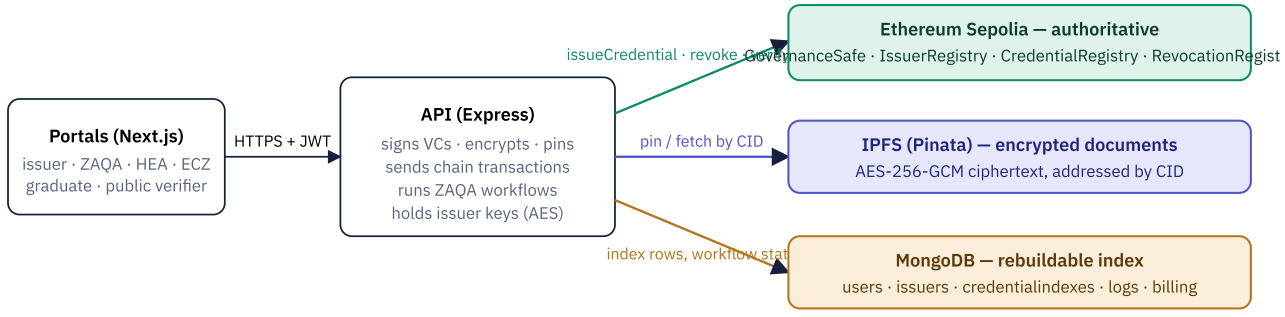
IPFS via Pinata

- The signed W3C Verifiable Credential, AES-256-GCM encrypted
- Institutions' uploaded accreditation documents
- Addressed by content (CID), so any change is a new CID

NON-AUTHORITATIVE INDEX

MongoDB (local)

- Accounts, roles, institution profiles, encrypted signing keys
- Credential index: names, qualification, year, QR payload, CID
- ZAQA validation state, suspensions, disputes, applications, billing
- Append-only event history per record

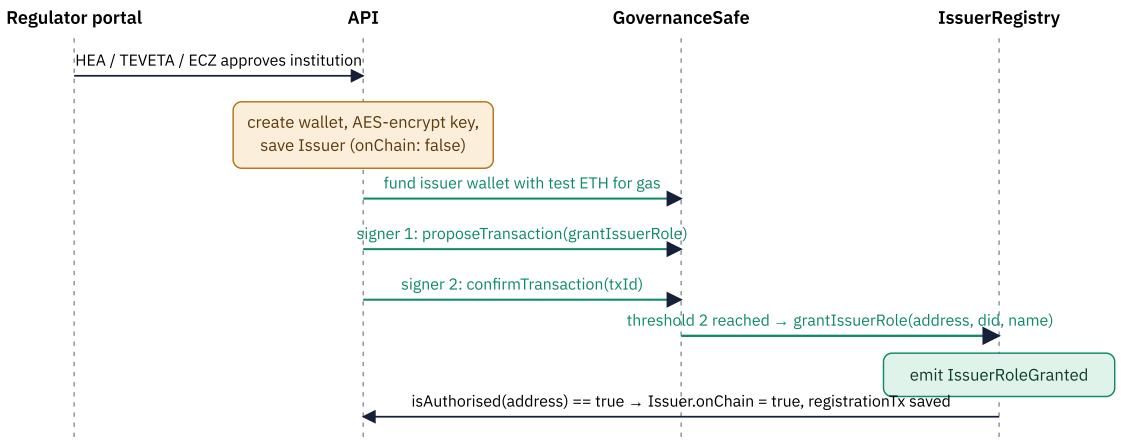


Every write goes through the API. Portals never talk to the chain directly except in MetaMask mode, where the institution's browser sends the anchoring transaction itself (section 3). The verifier path reads the chain through the API with a zero-gas call.

> **Why this ranking matters.** Deleting the MongoDB database loses names and workflow state, but every anchored credential can still be re-verified, and the index can be rebuilt from the `CredentialIssued` events plus the IPFS documents. Editing a MongoDB row cannot make a tampered certificate pass, because the verifier recomputes and compares hashes against the chain.

2 Authorising an institution to issue

Before an institution can anchor anything, its wallet address must hold `ISSUER_ROLE` in the `IssuerRegistry` contract. Only the `GovernanceSafe` can grant that role, and the Safe needs two of its three regulator signatures. In the deployed prototype the API holds signer keys 1 and 2 and performs both steps.



Two regulator signatures, one on-chain role grant. The institution's record is saved first with `onChain: false`, so a failed transaction can be retried without creating a duplicate wallet. Self-registered institutions sit at `heaStatus: pending` and only reach this sequence when their regulator approves them.

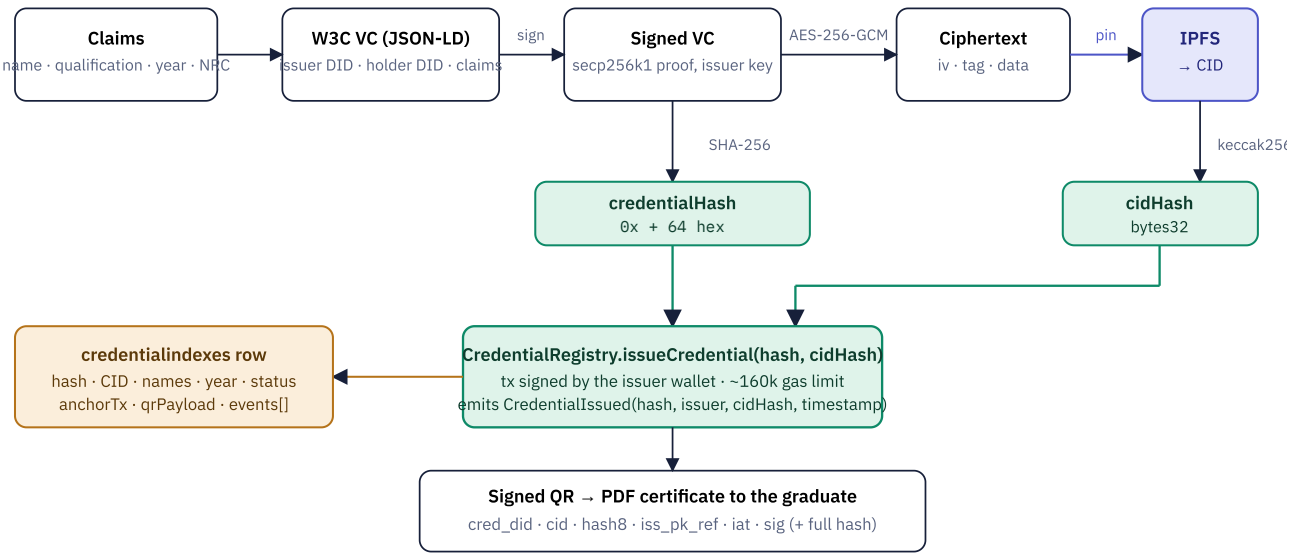
What the registry stores per issuer

did	keccak256 of the string `did:ethr:0x…`, stored as bytes32
institution	name truncated to 31 bytes, stored as bytes32 (a label, not the legal record)
registeredAt	block timestamp
active	true until `revokeIssuerRole` is executed through the Safe

Two signing modes exist. In **server mode** the API generates the wallet, and that same key later signs the anchoring transaction. In **MetaMask mode** the institution's own MetaMask address receives the role, and the API only ever signs the VC and QR with a separate server-held key. Either way the key that signs credentials at rest is AES-256-GCM encrypted in MongoDB.

3 Issuing a credential

Issuance is a fixed pipeline in `issuance.service.js`, shared by the institution portal, the ECZ certificate register and the graduate application workflow, so every credential in the system has exactly the same shape.



Only two 32-byte values reach the chain. The credential hash commits to the whole signed document, and the CID hash commits to the encrypted copy on IPFS. Names, the NRC and the qualification title never leave the API and MongoDB in plain form, and IPFS only ever holds ciphertext.

1 **Gate checks.** The institution must be regulator-approved, the qualification must match an accredited programme when a programme register exists, and a duplicate for the same person, programme and year is refused (409). An in-memory lock stops a double-click from reaching the chain twice.

2 **Build the VC.** `buildCredential` creates a W3C Verifiable Credential with the issuer DID, the holder DID (an existing graduate account's DID, otherwise a fresh `did:ethr`), and the claims.

3 **Sign it.** `signCredential` attaches an `EcdsaSecp256k1Signature2019` proof made with the institution's decrypted signing key.

4 **Hash it.** `sha256(JSON.stringify(signedVC))` is the credential hash. This is the identifier printed inside the QR and used everywhere else.

5 **Encrypt and pin.** The signed VC is AES-256-GCM encrypted with the server key (random 12-byte IV per credential) and uploaded to Pinata as JSON. Pinata returns the CID;

`keccak256(CID)` becomes the second on-chain value.

6 **Anchor.** Server mode: the API signs `issueCredential(hash, cidHash)` with the issuer wallet and waits for the receipt. MetaMask mode: `POST /prepare` returns hash, cidHash and the contract address; the browser sends the transaction; `POST /finalize` re-reads the chain and only activates the record when `verifyCredential` already answers VERIFIED.

7 **Index and deliver.** The MongoDB row is written with status `active` (or `pending` until MetaMask finalises), the transaction hash, and the signed QR payload. The graduate can download a PDF with that QR from their portal.

The signed QR

The QR is the offline credential. It carries six fields and a signature over five of them (`cred_did|cid|hash8|iss_pk_ref|iat`), plus the full credential hash for the online path. A verifier who has cached issuer public keys can check the signature with no network at all.

```
{ "cred_did": "did:ethr:0x9A…",  "cid": "bafkrei…",  "hash8": "3f9c1b7e2a04d6c8",
  "iss_pk_ref": "did:ethr:0x5B…#keys-1",  "iat": 1757900000,
  "sig": "0x3045…",  "hash": "0x3f9c1b7e2a04d6c8…" }
```

4 The on-chain record, field by field

Each anchored credential occupies three storage slots in `CredentialRegistry`, keyed by the credential hash. Nothing in it identifies the graduate.

FIELD	TYPE	SET BY	MEANING
issuer	address	`msg.sender`	Wallet that anchored the credential. Must hold ISSUER_ROLE at that moment.
issuedAt	uint64	`block.timestamp`	Anchoring time. Shown on verification as "Issued".
cidHash	bytes32	API	keccak256 of the IPFS CID of the encrypted VC.
issuerDID	bytes32	IssuerRegistry	Copied from the registry at issue time, so a later change of DID cannot rewrite history.

The contract refuses a hash that is already present, so a credential can only be anchored once. Revocation lives in a separate `RevocationRegistry` row: reason code 1 to 5, timestamp, revoking address, and optionally the hash of a corrected replacement.

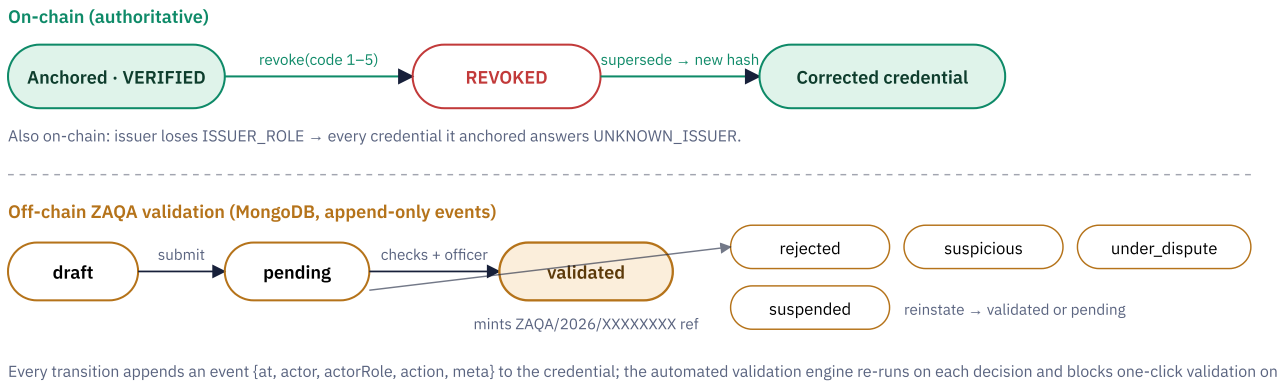
`verifyCredential` is a `view` function that combines both registries plus the issuer's current authorisation into one of four answers.

CONTRACT	SEPOLIA ADDRESS	WRITES HAPPEN WHEN
GovernanceSafe	`0x39aea70a…f990`	A regulator approves an institution (propose, confirm, auto-execute)
IssuerRegistry	`0x99B41d97…73ED`	Executed by the Safe: grant or revoke ISSUER_ROLE
CredentialRegistry	`0x785c5204…B8D2`	Every issuance and every corrected reissue
RevocationRegistry	`0xB23E0922…8263`	Institution or ZAQA revokes; supersession sets reason 1 and links the new hash

Approximate cost on Sepolia today: about 0.01 test ETH per institution registration and about 0.002 per anchor. The ZAQA funder wallet tops up new issuer wallets automatically so institutions never handle gas.

5 Validation and lifecycle: what is on-chain and what is not

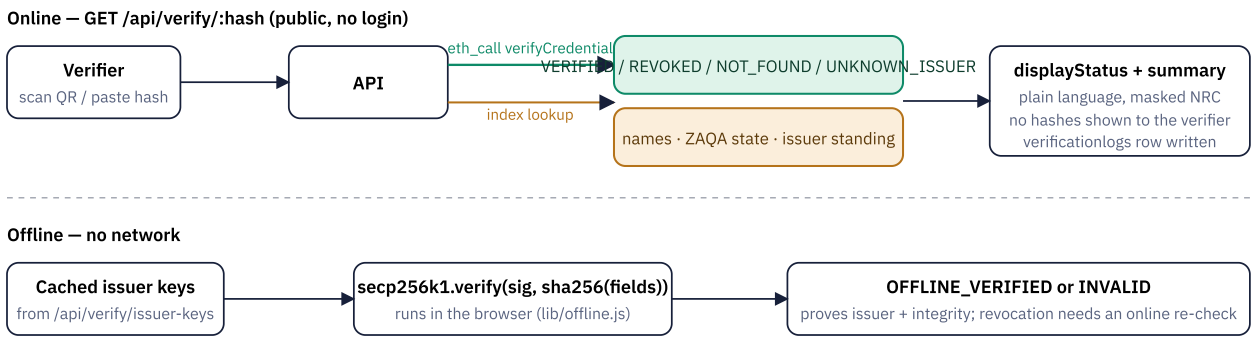
The chain answers one question: is this exact document genuine and not revoked. Everything about national recognition is a ZAQA decision recorded off-chain, and the verify response folds both together.



Two tracks, one answer. The verify response reports the on-chain status first (REVOKED, NOT_FOUND and UNKNOWN_ISSUER always win), then the ZAQA state: `validated` shows as Verified, `pending` as "Awaiting ZAQA validation", and so on. Suspension is purely off-chain: the anchor stays VERIFIED but the certificate download is blocked.

A ZAQA revocation is genuinely on-chain: the officer's decision makes the API sign `RevocationRegistry.revoke` with the custodied issuer key, so third parties see REVOKED without trusting BBACVS's database. A correction never edits the old record. The institution issues a new credential through the same pipeline, then the old hash is revoked with reason 1 (administrative error) and both index rows link to each other.

6 Verifying: online and offline



The online path never trusts the index for validity. The chain call decides the headline; MongoDB only adds human-readable context and the regulator's standing. The offline path proves that an authorised institution signed this exact QR, which is what a border post or employer with no signal needs first.

The ZAQA verification certificate

Once a credential is on-chain VERIFIED and ZAQA-validated, anyone can download the "Certificate of Verification and Evaluation of Qualification" PDF. It prints no hash, CID or wallet address at all. A second signed QR, made with ZAQA's own key (`did:zaqa#keys-1`) over the ZAQA reference and holder DID, carries the technical data so the certificate itself can be checked the same way.

7 Where each piece of data lives

DATA	STORE	NOTES
Credential hash (SHA-256 of signed VC)	Chain Index QR	Chain copy is authoritative; index and QR carry it for lookup.
IPFS CID of the encrypted VC	Index QR · Chain (keccak only)	The chain holds a hash of the CID, enough to detect a swapped document.
Signed VC document	IPFS	AES-256-GCM ciphertext; readable only with the server key.
Issuer wallet, DID, authorisation, revocation of role	Chain Index	Chain decides; index mirrors as `onChain`, `registrationTx`.
Issuer signing private key	Index	AES-encrypted at rest in the issuers collection. Never on chain or IPFS.
Graduate name, NRC, qualification title, year, ZQF level	Index IPFS (encrypted)	Never on chain. NRC is masked in every public response.
Revocation with reason code, supersession link	Chain Index	Chain is authoritative; index adds `supersededBy` / `supersedes` both ways.
ZAQA validation state, ZAQA reference, validation report	Index	Regulatory decision, off-chain by design, with append-only events.
Suspension, disputes, correction requests	Index	Off-chain; suspension blocks certificate download but leaves the anchor VERIFIED.
Programmes, qualifications register, NQF levels	Index	Configuration data used by the validation engine's checks.
Applications, evidence uploads, accreditation documents	Index IPFS	Uploaded files are pinned to IPFS; the CID is kept in the index.
Invoices, payments, receipts, notifications, activity log	Index	Operational records only.
Verification attempts	Index	`verificationlogs`: hash, result, mode, latency. No PII.